Macroeconomics I — MPE Insper

Practice list: The New Keynesian AS–AD Model — qualifier level (Lista 7)

Instructions. Multiple choice: mark the single correct option (no penalty for wrong answers). *True/False*: every 2 wrong items cancel 1 right one (score = $\max\{0, \text{right} - \text{wrong}/2\}\times$ item value); blank items do not count. Discursive questions: show every step. Parameters are symbols with the stated admissible ranges; answers are closed forms, conditions, signs and limits.

Notation (Benigno 2015). Lower-case letters are log deviations from the steady state; bars denote the long run. Households: $u(C) - v(L) + \beta\{u(\bar{C}) - v(\bar{L})\}$, $u = C^{1-\tilde{\sigma}^{-1}}/(1-\tilde{\sigma}^{-1})$, $v = L^{1+\eta}/(1+\eta)$, $\rho \equiv -\ln\beta$. Resources: $y = s_c c + g$, $\sigma \equiv \tilde{\sigma} s_c$. Firms: demand $Y(j) = (P(j)/P)^{-\theta}(C+G)$, technology $Y(j) = AL(j)$; a fraction α of prices is preset at P^e. Parameters: $0 < \alpha < 1$, $\sigma, \eta, \rho > 0$, $\theta > 1$; unless stated, $p^e = \bar{p} = 0$, $g = \bar{g} = 0$ and no taxes.

Block 1 — Foundations: AD, AS, natural and efficient output, taxes

(Benigno §§1–5 • 25 points)

Question 1. (10 points) Objective items.

a) (3 points) The slope κ of the AS curve $p - p^e = \kappa(y - y_n)$, in terms of the primitives, is:

A. $(1-\alpha)(\sigma^{-1}+\eta)$

B. $\alpha(\sigma^{-1}+\eta)/(1-\alpha)$

C. $(1-\alpha)(\sigma+\eta)$

D. $(1-\alpha)(\sigma^{-1}+\eta)/\alpha$

b) (3 points) *AS–AD × taxation.* Short-run productivity rises ($a > 0$) and a payroll tax raises the (log) markup by $\mu > 0$, both temporarily. The changes in natural and efficient output, $(\Delta y_n,\ \Delta y_e)$, are:

A. $((a-\mu)/(\sigma^{-1}+\eta),\ a/(\sigma^{-1}+\eta))$

B. $([(1+\eta)a-\mu]/(\sigma^{-1}+\eta),\ (1+\eta)a/(\sigma^{-1}+\eta))$

C. $([(1+\eta)a-\mu]/(\sigma^{-1}+\eta),\ [(1+\eta)a-\mu]/(\sigma^{-1}+\eta))$

D. $(a/(\sigma^{-1}+\eta),\ a/(\sigma^{-1}+\eta))$

c) (2 points, T or F) Because θ sets the markup, a higher demand elasticity θ makes the AS curve steeper.

d) (2 points, T or F) *AS–AD × fiscal news.* Holding i, $\bar{p}$ and p^e fixed, news that long-run government spending will be higher ($\bar{g}\uparrow$, g unchanged) lowers current output.

Question 2. (15 points) Discursive. *AS–AD foundations × taxation × supply shocks.* Build the model from its primitives (no taxes unless stated).

a) (3 points) From the household problem with budget $PC + \frac{\bar{P}\bar{C}}{1+i} = WL + \frac{\bar{W}\bar{L}}{1+i} + T$, derive the Euler equation and the labor-supply condition; log-linearize and combine with $y = s_c c + g$ to obtain the AD curve. Give its slope in the (y, p) plane, explain it, and sign the effect of each shifter.

Solution:

b) (3 points) From the firm's problem derive the optimal flexible price, natural output and the AS curve with a fraction α of preset prices.

Solution:

c) (3 points) Derive efficient output from the planner's problem and compare it with natural output. Show how labor, payroll, sales and consumption taxes affect each, and find the sales subsidy that makes the steady state efficient. How does it depend on θ?

Solution:

d) (3 points) Start from an efficient steady state ($y_n = y_e = 0$, as in (c)). An oil shock raises the markup by $\mu > 0$ and lowers productivity, $a < 0$, for the short run only. Find the output gap in terms of the natural real rate and the policy rate. Characterize y, $p - p^e$ and $y - y_e$ if the bank keeps $i = \rho$; find the i that delivers $p = p^e$ and its outcome. Draw the AS–AD diagram (old and new AS, AD, both equilibria, y_n and y_e after the shock).

Solution:

e) (3 points) *Parameter analysis.* For the markup shock alone ($\mu > 0$, $a = 0$, $i = \rho$), find y, $p - p^e$ and the natural gap $y - y_n$ in closed form, sign their responses to α, σ, η and μ with proofs in a table, give the limits $\alpha \to 0$ and $\alpha \to 1$, and read the results.

Solution:

Block 2 — Policy: optimal rules, the zero bound and household debt

(Benigno §§6–12 • 25 points)

Question 3. (10 points) Objective items. Ranges: $\sigma, \eta, \kappa, \Delta > 0$; the long run is unchanged unless stated.

a) (3 points) *Monetary × fiscal policy.* A temporary cut Δ in the consumption tax (like the 2009 IPI cut on cars: $\tau_c = -\Delta$, $\bar{\tau}_c = 0$), entering the markup one for one, changes natural output y_n and the natural real rate r_n (the rate at which demand equals y_n) by, respectively:

A. $\left(\frac{\Delta}{\sigma^{-1}+\eta},\ \frac{\eta\Delta}{\sigma^{-1}+\eta}\right)$

B. $\left(\frac{\Delta}{\sigma^{-1}+\eta},\ -\frac{\sigma^{-1}\Delta}{\sigma^{-1}+\eta}\right)$

C. $\left(\frac{(1+\eta)\Delta}{\sigma^{-1}+\eta},\ \frac{(1-\sigma^{-1})\eta\,\Delta}{\sigma^{-1}+\eta}\right)$

D. $\left(\frac{\Delta}{\sigma^{-1}+\eta},\ \frac{(2\sigma^{-1}+\eta)\Delta}{\sigma^{-1}+\eta}\right)$

b) (3 points) *Fiscal multipliers.* Let $\Omega \equiv (\sigma^{-1} + \eta)(1 + \sigma\kappa)$. With i, $\bar{p}$ and p^e fixed, a temporary rise in g ($\bar{g}$ unchanged, lump-sum financed) changes output and the gap $y - y_n$, per unit of g, by:

A. $\left(\frac{\sigma^{-1}+\eta}{\Omega},\ \frac{\sigma^{-1}+\eta}{\Omega}\right)$

B. $\left(\frac{\sigma^{-1}+\eta+\kappa}{\Omega},\ \frac{\sigma^{-1}+\eta+\kappa}{\Omega}\right)$

C. $\left(\frac{\sigma^{-1}+\kappa}{\Omega},\ 0\right)$

D. $\left(\frac{\sigma^{-1}+\eta+\kappa}{\Omega},\ \frac{\eta}{\Omega}\right)$

c) (2 points, T or F) At the zero lower bound with $r_n < 0$, a credible commitment to a long-run price level higher by $-r_n$ restores $y = y_n$ and $p = p^e$, and the size of the commitment needed does not depend on the degree of price rigidity.

d) (2 points, T or F) *AS–AD × household debt.* In the borrower–saver economy of Question 4(d), with constant inflation between the short and the long run, more price flexibility reduces the output cost of a deleveraging shock.

Question 4. (15 points) Discursive. *Optimal policy × zero lower bound × household debt.* Ranges: $\sigma, \kappa, \rho > 0$, $\eta \geq 0$, $\theta > 1$; in (d)–(e), $\chi \in (0, 1)$, $d_0 > 0$, $\beta \in (0, 1)$. Shocks are temporary.

a) (3 points) The bank minimizes Benigno's welfare loss $L = \frac{1}{2}(y - y_e)^2 + \frac{\theta}{2\kappa}(p - p^e)^2$ (eq. 33) subject to the AS. Find the optimal relation between the efficient gap and price surprises, describe it in the (y, p) plane, and prove that shocks that leave μ unchanged create no trade-off.

Solution:

b) (3 points) A temporary cost-push shock raises the markup by $\mu > 0$ and moves productivity by a (either sign; Question 2(d) is the case $a < 0$); $\bar{y}_n = 0$, $\bar{p} = p^e$, $g = \bar{g}$. Find the optimal y, $p - p^e$ and i; compare them and the loss with strict price stability; say which part of the shock i offsets one for one. Show that no Taylor rule $i = \bar{\imath} + \psi_y(y - y_e) + \psi_\pi(p - p^e)$, $\psi_y, \psi_\pi \geq 0$, with a constant intercept $\bar{\imath}$ (a policy constant, not a long-run value) implements the optimum under all productivity shocks.

Solution:

c) (3 points) Pessimism: long-run productivity is expected to fall, $\bar{a} < 0$, with $a = \mu = 0$ and $g = \bar{g}$. Find when the zero bound binds, the outcome at $i = 0$ with $\bar{p} = p^e$, and the $\bar{p}$ that restores the optimum; give the limits as $\kappa \to \infty$ and $\kappa \to 0$, and discuss credibility.

Solution:

d) (3 points) With borrowers and savers (share χ of savers, initial debt d_0), Benigno's AD is

$$y = \bar{y}_n - \varphi[i - (\bar{p}_n - p) - \rho] + \tfrac{1}{\chi}(g - \bar{g}) + \tfrac{1-\chi}{\chi}[\hat{d} + d_0(p - p^e)], \qquad \varphi \equiv \frac{\sigma\chi + (1-\chi)d_0\beta}{\chi},$$

where $\bar{p}_n$ is the long-run price level ($\bar{p}$ above) and $\hat{d}$ the change in the debt limit. Find dy/dp when $\bar{p}_n$ is anchored and when inflation $\bar{p}_n - p$ is constant, naming the two channels. Assuming the anchored AD slopes down and, under constant inflation, the AD is steeper than the AS, derive the spending multipliers m_A, m_C (i fixed), order them against $1/\chi$, and find when $m_A > 1$.

Solution:

e) (3 points) *Parameter analysis.* A deleveraging shock lowers the debt limit, $\hat{d} < 0$, with i, g fixed and the economy initially at $y = y_n = 0$, $p = p^e$. Find each regime's output response and the stability bound; sign both responses and both multipliers of (d) against κ, d_0, χ, σ (given κ) in a table, with proofs; give the limits in κ and the knife-edges; draw both responses against κ and read them.

Solution:

Block 3 — Integrative: a pandemic in the AS–AD model

(Benigno §§4–11; Kurlat chs. 6–7 • 25 points)

Question 5. (10 points) Objective items.

a) (3 points) *Expectations × AS–AD.* To fight a pandemic slump with i fixed, the central bank credibly raises the long-run price level by $d\bar{p} > 0$; the preset prices p^e are predetermined. Current output changes by:

A. $dy = \sigma\, d\bar{p}$

B. $dy = \frac{\sigma}{1+\sigma\kappa}\, d\bar{p}$

C. $dy = d\bar{p}$

D. $dy = \frac{\sigma\kappa}{1+\sigma\kappa}\, d\bar{p}$

b) (3 points) *Growth × AS–AD.* Productivity rises permanently, $a = \bar{a} > 0$ (say, lasting gains from remote work), while the bank keeps $i = \rho$ and $\bar{p} = p^e$. The new equilibrium $(y,\ p - p^e)$ is:

A. $(0,\ -\frac{\kappa(1+\eta)a}{\sigma^{-1}+\eta})$

B. $(\frac{(1+\eta)a}{\sigma^{-1}+\eta},\ -\frac{\kappa(1+\eta)a}{\sigma^{-1}+\eta})$

C. $(\frac{(1+\eta)a}{\sigma^{-1}+\eta},\ 0)$

D. $(a,\ 0)$

c) (2 points, T or F) *Consumption theory × AS–AD.* Hold i and the long-run price level fixed. A lump-sum transfer financed by public debt (such as Brazil's 2020 *auxílio emergencial*) leaves output unchanged in Benigno's representative-agent model, whereas in his borrower–saver model (eq. (32), stable equilibrium) a debt-financed transfer to the borrowers raises output.

d) (2 points, T or F) *Labor × AS–AD.* Okun's law maps gaps into unemployment, $u - u_n = -\gamma(y - y_n)$ with $\gamma > 0$. After the oil shock of Question 2(d) (short run only: the markup rises, $\mu > 0$, and productivity falls, $a < 0$), a bank that keeps $i = \rho$ and $\bar{p} = p^e$ leaves unemployment below its natural rate even though output falls.

Question 6. (15 points) Discursive. *Integrative: supply shocks × household debt × fiscal transfers × optimal policy.* A stylized pandemic hits the borrower–saver economy of Benigno §10: a share $\chi \in (0,1)$ of savers with discount factor $\beta \in (0,1)$, $\rho = -\ln\beta$; borrowers at their debt limit, with initial debt $d_0 > 0$; utility $u(C_j) = 1 - e^{-zC_j}$, $z > 0$, with $\sigma \equiv 1/(z\tilde{Y})$ ($\tilde{Y}$: steady-state output), and $v(L_j) = L_j^{1+\eta}/(1+\eta)$; Cobb–Douglas indices $L = L_s^{\chi}L_b^{1-\chi}$ and $W = W_s^{\chi}W_b^{1-\chi}$. For the short run only, productivity falls ($a < 0$) and the markup rises ($\mu > 0$); households expect lower long-run productivity ($\bar{a} < 0$); banks tighten the debt limit ($\hat{d} < 0$). Long-run prices stay anchored, $\bar{p}_n = p^e = 0$; initially $g = \bar{g} = 0$ and $\tau_b = \bar{\tau}_b = 0$, where τ_b is the lump-sum tax on each borrower (debt and taxes in shares of $\tilde{Y}$). Benigno's AD (32) reads

$$y = \bar{y}_n - \varphi\big[i - (\bar{p}_n - p) - \rho\big] + \tfrac{1}{\chi}\big[(g - \bar{g}) - (1-\chi)(\tau_b - \bar{\tau}_b)\big] + \tfrac{1-\chi}{\chi}\big[\hat{d} + d_0(p - p^e)\big],$$

with $\varphi \equiv [\sigma\chi + (1-\chi)d_0\beta]/\chi$.

a) (3 points) *Supply side.* Show that the AS curve $p - p^e = \kappa(y - y_n)$, natural output and efficient output do not depend on how consumption is split between the two types. Find y_n, y_e and $\bar{y}_n$ after the shocks, and say which shock moves which of them and which curve it shifts.

Solution:

b) (3 points) *Demand and equilibrium.* Find the slope of the AD curve, the condition under which it slopes down and the channels behind it; the equilibrium y and p; the natural and efficient

gaps as functions of i; the rate i_n that closes the natural gap; and the condition under which the zero lower bound $i \geq 0$ rules i_n out.

Solution:

c) (3 points) *Fiscal transfers.* The government pays Δ to each borrower ($\tau_b = -\Delta$, $\bar{\tau}_b$ unchanged) and taxes the savers in the long run to repay the debt. From the borrowers' budget constraint, derive the output effect per unit of aggregate transfer; find the transfer Δ_e that closes the efficient gap at $i = \rho$; find the effect of a transfer to savers financed the same way; and study $\chi \to 1$.

Solution:

d) (3 points) *Policy mix.* The bank follows the targeting rule of Question 4(a). Compare (i) cutting i subject to $i \geq 0$, (ii) transfers alone with $i = \rho$, and (iii) both. When the bound binds, find the transfer Δ_{zlb} that implements the optimum at $i = 0$ and the welfare cost of not paying it. Draw the AS–AD diagram: old and new AS; AD before the shocks, after them at $i = 0$, and after the policy; the equilibria.

Solution:

e) (3 points) *Parameter analysis.* Sign, with proofs, the effects of χ, d_0 and κ (through α) on the transfer multiplier and on Δ_{zlb}, and collect them in a table; compare with the multiplier when inflation between the short and the long run is constant instead; give the limits and the stability conditions in both regimes; and read the results for Brazil's 2020 emergency transfers.

Solution: